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United States Patent Application Publication

20250282581

Kind Code

A1

Publication Date

September 11, 2025

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PIPE SLING

Abstract

A pipe sling includes a fabric body section having a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section. A first set of securement devices are affixed to a first segment of the fabric body section defined by the second axis of symmetry. A second set of securement devices are affixed to a second segment of the fabric body section defined by the second axis of symmetry. At least one reinforcement element is positioned along the first axis of symmetry and extends from a first end of the fabric body section to a second end of the fabric body section.

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Appl. No.: 18/601358

Filed: March 11, 2024

Publication Classification

Int. Cl.: B66C1/18 (20060101); F16L1/06 (20060101); F16L3/137 (20060101); F16L3/26 (20060101)

U.S. Cl.:

CPC B66C1/18 (20130101); F16L1/06 (20130101); F16L3/137 (20130101); F16L3/26 (20130101);

Background/Summary

BACKGROUND

[0001] Damaged pipe removal is often a tedious process to carry out. Specifically, pipes that have questionable structural integrity, such as pipes that are corroded, thin-walled, and/or eroded, are delicate and thus become very cumbersome to move or transport. A common traditional means of removing a pipe from usage includes the choking of nylon slings from a crane at each end of the pipe with nothing along the length to brace the pipe in the case of a failure. When a long length of piping has corroded past its own structural limitations, shorter cuts of the pipe may have to be made to reduce the overall length of the pipe in order to support itself. This can result in more work being involved in a project such as, for example, additional cuts, additional stripping of insulation (if required), additional lifts utilizing the crane, and additional asbestos abatement work. This work will raise the overall cost of the project and increase overall exposure hours.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] For a more complete understanding of the features and advantages of the present disclosure, reference is now made to the detailed description along with the accompanying figures in which corresponding numerals in the different figures refer to corresponding parts and in which:

[0003] FIG. 1A is an illustration of an exterior view of a pipe sling in an open configuration in accordance with certain embodiments of the present disclosure;

[0004] FIG. 1B is a photo of an exterior view of a pipe sling in an open configuration in accordance with certain embodiments of the present disclosure;

[0005] FIG. 1C is an illustration of a perspective view of the pipe sling of FIG. 1A in a secured position in accordance with certain embodiments of the present disclosure; and

[0006] FIG. 2 is an illustration of a perspective view of the pipe sling of FIG. 1A being suspended from a lifting device while supporting a pipe in accordance with certain embodiments of the present disclosure.

[0007] The illustrated figures are only exemplary and are not intended to assert or imply any limitation with regard to the environment, architecture, design, or process in which different examples may be implemented.

DETAILED DESCRIPTION

[0008] The present invention relates generally to slings, and more particularly to slings for lifting pipes out of the ground. The term “safe working load” or “SWL” is used herein to describe a maximum safe force that a piece of lifting equipment, lifting device, or accessory can exert to lift, suspend, or lower, a given mass without fear of breaking.

[0009] By virtue of utilizing embodiments of a pipe sling as disclosed, in regards to a pipe, additional structural support can be added, capture or encapsulation support can be provided in the event of failure, and average overall length (of 12 feet from cut to cut) can be maintained. This can reduce the overall scope of the work, reducing the cost of the project and make the project safer by reducing exposure hours and preventing lift failures. Alternatively, certain client policies are to remove insulation from carbon steel piping and perform an inspection of pipe before removing, regardless of the integrity. Utilizing the disclosed embodiments of the sling can eliminate the inspection/removal steps of the project, shortening the duration and reducing the cost of the project overall.

[0010] While the making and using of various embodiments of the present disclosure are discussed in detail below, it should be appreciated that the present disclosure provides many applicable inventive concepts, which can be embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative and do not delimit the scope of the present

disclosure. In the interest of clarity, not all features of an actual implementation may be described in the present disclosure.

[0011] Unless otherwise indicated, all numbers expressing quantities of components, properties such as molecular weight, reaction conditions, and so forth used in the present specification and associated claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the examples of the present invention. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claim, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. It should be noted that when “about” is at the beginning of a numerical list, “about” modifies each number of the numerical list. Further, in some numerical listings of ranges some lower limits listed may be greater than some upper limits listed. One skilled in the art will recognize that the selected subset will require the selection of an upper limit in excess of the selected lower limit. Further, although voltage signals discussed herein are discussed as positive voltages, a negative voltage signal or differential voltage signal may be generated, used, or detected without departing from the scope of the present invention.

[0012] Presented herein is a pipe sling. The pipe sling comprises a fabric body section, including a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section. A first set of securement devices are affixed to a first segment of the fabric body section defined by the second axis of symmetry. A second set of securement devices are affixed to a second segment of the fabric body section defined by the second axis of symmetry. At least one reinforcement element is positioned along the first axis of symmetry and extends from a first end of the fabric body section to a second end of the fabric body section.

[0013] Also presented herein is a system for transporting pipes comprising a lifting device and a pipe sling suspended from the lifting device and wrapped around a diameter of a pipe. The pipe sling comprises a fabric body section comprising a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section. A first set of securement devices affixed to a first segment of the fabric body section are defined by the second axis of symmetry, where a first quantity of the first set of securement devices are secured to the lifting device and a second quantity of the first set of securement devices are wrapped around and secured to the fabric body section and the pipe. A second set of securement devices affixed to a second segment of the fabric body section are defined by the second axis of symmetry, where a first quantity of the second set of securement devices are removably affixed to the lifting device and a second quantity of the second set of securement devices are wrapped around and secured to the fabric body section and the pipe. At least one reinforcement element positioned along the first axis of symmetry extends from a first end of the fabric body section to a second end of the fabric body section. The at least one reinforcement element is configured to reduce sag of the fabric body section when the pipe sling is wrapped around the pipe while secured to the lifting device.

[0014] Also presented herein is a pipe sling for supporting a pipe. The pipe sling comprises a nylon canvas fabric body section comprising a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section, where the width of the fabric body section ranges from 24 inches to 36 inches. A first set of securement devices affixed to a first segment of the fabric body section are defined by the second axis of symmetry and a second set of securement devices affixed to a second segment of the fabric body section are defined by the second axis of symmetry. Each of the first set of securement devices and each of the second set of securement devices extend across the width of the fabric body section and are stitched to the fabric body section along the first axis of symmetry, where the first

set of securement devices and the second set of securement devices include at least four eye to eye nylon lifting straps spaced between at least four ratchet straps. At least one reinforcement element comprising at least one metal segment extends along an entirety of the first axis of symmetry from a first end of the fabric body section to a second end of the fabric body section, where the at least one reinforcement element is configured to reduce sag of the fabric body section when the pipe sling is being lifted with a pipe. A safe working load (SWL) of the pipe sling, when lifting the pipe, ranges from 3,000 pounds to 4,500 pounds.

[0015] FIG. 1A is an illustration of an exterior view of a pipe sling **100** in an open position in accordance with certain embodiments of the present disclosure. Pipe sling **100** comprises a fabric body section **110**, where, in embodiments, is a nylon canvas fabric. The fabric body section **110** includes an exterior surface area, an interior surface area (for contact with a pipe), a first axis of symmetry **112** extending along a length of the fabric body section **110**, and a second axis of symmetry **114** extending along a width of the fabric body section **110**. First axis of symmetry **112** may segment fabric body section **110** into first and second (upper and lower) segments **115,116**, while second axis of symmetry **114** may also segment fabric body section **110** into first and second (left and right) segments **117,118**. A first set of securement devices **122** are affixed to segment **117** defined by second axis of symmetry **114**. The first set of securement devices **122** include a pair of lifting straps **123** positioned between a pair of ratchet straps **124**. A second set of securement devices **126** symmetrical to the first set of securement devices **122** (in relation to the second axis of symmetry **114**) are affixed to segment **118** and also include a pair of lifting straps **123** positioned between a pair of ratchet straps **124**. The (four) ratchet straps **124** are positioned to effectively tighten fabric sling body section **110** of pipe sling **100** around a pipe while minimizing slippage and securing insulation (if needed) around the pipe. This may prevent/reduce the potential of the pipe slipping out of the pipe sling **100**/being dropped.

[0016] As further shown in FIG. 1A, each of the first set of securement devices **122** and each of the second set of securement devices **126** extend across the width of the fabric body section **110** and are secured (stitched) to exterior surface area of fabric body section **110** along the first axis of symmetry **112**. In order to keep the first and second set of securement devices **122,124** properly aligned across the width of the fabric body section **110**, each of the first set of securement devices **122** and each of the second set of securement devices **124** are positioned through a pair of guide loops **133**, where each of the pair of guide loops **133** are positioned equidistant from the first axis of symmetry **112**. In certain embodiments, first and second guide loops of a pair of guide loops **133** of each of the securement devices of the first set and second set of securement devices are positioned 12 inches from a second guide loop of each of the pair of guide loops **133** of the first set of securement devices **124** and the second set of securement devices **126**. By virtue of the loop configurations, a secure foundation is provided to pipes being transported by pipe sling **100**.

[0017] In an embodiment, each of the first set of securement devices **122** and each of the second set of securement devices **126** are secured (stitched) to fabric body section **110** adjacent edges of first and second (upper and lower) segments **115,116**.

[0018] When a pipe **200** is positioned on/within pipe sling **100**, such as the embodiment shown in FIG. 1B, lifting straps **123** are configured to attach to a lifting device (or to an adapter configured to attach to a lifting device) for transport of the pipe **200** while ratchet straps **124** are configured to wrap around pipe **200** and secure pipe **200** within fabric body section **110**. In order to reduce sag/sagging of the fabric body section **110** when pipe sling **100** is being lifted with a pipe **200**, at least one reinforcement element **130** is positioned along the first axis of symmetry **112**. The at least one reinforcement element **130**, as shown, includes a one inch metal segment and extends from a first end **132** of the fabric body section **110** to a second end **134** of the fabric body section **110**. Additionally, in embodiments, the at least one reinforcement element **130** is sewn between fabric body section **110** and another piece of fabric (such as, for example, a webbing). In an embodiment, the at least one reinforcement element **130** includes two one inch metal segments that are sewn

between fabric body section **110** and another piece of fabric that extend from a first end **132** of the fabric body section **110** to a second end **134** of the fabric body section **110**. In a further embodiment, the at least one reinforcement element **130** comprises carbon steel.

[0019] In an embodiment, in the case where the at least one reinforcement element **130** is sewn between fabric body section **110** and another piece of fabric (such as, for example, a webbing), multiple pieces of webbing may be utilized in order to securely encapsulate one or more reinforcement elements **130** to fabric body section **110**.

[0020] FIG. **1C** is an illustration of a perspective view of the pipe sling **100** of FIG. **1A** in a secured position accordance with certain embodiments of the present disclosure. As shown, pipe sling **100** is wrapped around and secured to a pipe **200**. In order to secure pipe sling **100** to pipe **200**, a plurality of ratchet straps **124** (as shown, four ratchet straps) are positioned equidistantly from second axis of symmetry **114**. When pipe **200** is positioned within pipe sling **100**, strap portions of ratchet straps **124** are wrapped around pipe **200** and pipe sling **100**. Ends of the strap portions are fed through ratchet portions of the ratchet straps **124** until the ratchet straps **124** are tightly secured to/around pipe **200** and pipe sling **100**.

[0021] As further shown, the two outermost lifting straps **123** are shown in an engageable position and are each configured to receive a transitional receiver device (such as, for example, a shackle) when in this position. The two innermost lifting straps **123** are shown in a disengageable position and folded over themselves. If additional support is needed toward the center of pipe sling **100**, the innermost lifting straps **123** can be engageable in a similar fashion as the two outermost lifting straps **123**.

[0022] FIG. **2** is an illustration of a perspective view of the pipe sling **100** of FIG. **1A** being suspended from a lifting device **210** (as shown, for example, a crane) while supporting a pipe **200** in accordance with certain embodiments of the present disclosure. As shown, a portion of the outermost lifting straps **123** are positioned around fabric body section **110** and pipe **200** while looped ends of the outermost lifting straps **123** are engaged with a pair of shackles **150** attached to upper lifting straps **160** (or, in other embodiments, wire rope) suspended from a shackle **150** secured to a crane hook **165** of lifting device **210** so that pipe sling **100** and pipe **200** are suspended from lifting device **210**. Prior to pipe sling **100** being suspended from lifting device **210**, ratchet straps **124** are tightened securely around pipe **200** to be moved/transported. In instances where pipe sling **100** is utilized to move/transport pipes having asbestos insulation, these pipes can be safely and securely moved/transported. Utilizing pipe sling **100** in the enclosed configuration as shown can eliminate a step of prepping an asbestos-lined pipe for transportation purposes. In an embodiment, lifting device **210** may comprise a come along device. As previously presented, if additional support is needed toward the center of pipe sling **100**, the innermost lifting straps **123** can be engageable in a similar fashion as the two outermost lifting straps **123**. It is noted that at least shackles **150** and upper lifting straps **160** may be referred to as “rigging” for attaching pipe sling **100** and pipe **200** to lifting device **210**. In embodiments, “rigging” may attach to the two innermost lifting straps **123** only, the two outermost lifting straps **123** only, or to all four (or more, in embodiments) lifting straps **123**. In the case where all lifting straps **123** are to be engaged, multiple sets of equipment “rigging” may be utilized to engage all lifting straps **123**. For example, a first set of equipment rigging may engage the two innermost lifting straps **123** and connect the two innermost lifting straps **123** to a first lifting device **210** while a second set of equipment rigging may engage the two outermost lifting straps **123** and connect the two outermost lifting straps **123** to a second lifting device **210**.

[0023] Generally, pipe sling **100** is configured to assist in the moving and/or transport of pipes having a diameter of ½ inch to 12 inches and a length of no greater than 12 feet. In order to move and/or transport the pipes, pipe sling **100**, in embodiments, may comprise a fabric body section **110** having a width ranging from 24 inches to 36 inches and a length of 10 feet.

[0024] In embodiments, four lifting straps **123** may be rigged (using rigging as described above, for

example) to a single lifting device **210** for the lifting of pipe sling **100** and a pipe. In other embodiments, a first set/pair of lifting straps **123** (one positioned on upper segment **115** and one positioned on lower segment **116**) may be rigged to a single lifting device **210** while the second set/pair of lifting straps **123** may be rigged to an additional lifting device **210** (or may be engaged by one or more individuals). The second set/pair of lifting straps **123** may be engaged in order to maneuver pipe sling **100** and the pipe, via the additional lifting device **210**, around obstacles that may be in the way/in the lifting path of pipe sling **100** and the pipe. The obstacles may include, but are not limited to: pipes, pipe rack structures, wires, and electrical cables. It is noted that the second set/pair of lifting straps **123** may engage a lifting device **210** before lifting of the pipe sling **100** and pipe or during the process of lifting if it is discovered that there are obstacles in the lifting path of pipe sling **100** and pipe. In additional embodiments, during the lifting process if/when obstacles get in the way of the lifting path of the pipe sling **100** and pipe, the first set/pair of lifting straps **123** may be disengaged from pipe sling **100** and pipe and the second set/pair of lifting straps **123** may be engaged to pipe sling **100** and pipe; this may enable maneuverability of the pipe sling **100** and pipe around the obstacles (via a lifting device **210**) and may maintain points of attachment to pipe sling **100**. Generally, additional tie-off points are created when pipe sling **100** includes four lifting straps **123**. In additional embodiments, the aforementioned rigging configurations may be carried out when there are more or less than four lifting straps **123**.

[0025] In an embodiment, a pipe sling **100** is provided that is configured to support a pipe having a diameter ranging from ½ inch to 6 inches. Pipe sling **200** comprise a nylon canvas fabric body section **110** including a first axis of symmetry **112** extending along a length (in embodiments, 10 feet) of the fabric body section **110** and a second axis of symmetry **114** extending along a width (in embodiments, 24 inches) of the fabric body section **110**. A first set of securement devices **122** are affixed to a first (left) segment **117** of the fabric body section **110** defined by the second axis of symmetry **114** and a second set of securement devices **126** affixed to a second (right) segment **118** of the fabric body section **110** defined by the second axis of symmetry **114**, where each of the first set of securement devices **122** and each of the second set of securement devices **126** extend across the width of the fabric body section **110** and are stitched to the fabric body section **110** along the first axis of symmetry **112**. The first set of securement devices **122** and the second set of securement devices **126** include four eye to eye nylon lifting straps **123** (in embodiments, 1 inch by 6 feet, 2 ply) spaced between four ratchet straps **124**. A reinforcement element **130** includes a 1 inch metal segment extending along an entirety of the first axis of symmetry **112** from a first end **132** of the fabric body section **110** to a second end **134** of the fabric body section **110**, where the reinforcement element **130** is configured to reduce sag of the fabric body section **110** when the pipe sling **100** is being lifted with a pipe. It is noted that the pipe sling **100** described above is certified to embody a safe working load (SWL)/rated capacity of 3,000 pounds. The certification is based at least on the following: Federal Specification RR-W-410D, ASME B30.9, API 9A, API 9B, API 2D, the Wire Rope Technical Board's "Wire Rope User's Manual", the Wire Rope Technical Board's "Wire Rope Sling User's Manual", and the Occupational Safety and Health Administration's "Industrial Slings".

[0026] In an embodiment, a pipe sling **100** is provided that is configured to support a pipe having a diameter ranging from 6 inches to 12 inches. Pipe sling **200** comprise a nylon canvas fabric body section **110** including a first axis of symmetry **112** extending along a length (in embodiments, 10 feet) of the fabric body section **110** and a second axis of symmetry **114** extending along a width (in embodiments, 36 inches) of the fabric body section **110**. A first set of securement devices **122** are affixed to a first (left) segment **117** of the fabric body section **110** defined by the second axis of symmetry **114** and a second set of securement devices **126** affixed to a second (right) segment **118** of the fabric body section **110** defined by the second axis of symmetry **114**, where each of the first set of securement devices **122** and each of the second set of securement devices **126** extend across the width of the fabric body section **110** and are stitched to the fabric body section **110** along the

first axis of symmetry **112**. The first set of securement devices **122** and the second set of securement devices **126** include four eye to eye nylon lifting straps **123** (in embodiments, 2 inches by 6 feet, 2 ply) spaced between five ratchet straps **124**. At least one reinforcement element **130** includes two 1 inch metal segments extending along an entirety of the first axis of symmetry **112** from a first end **132** of the fabric body section **110** to a second end **134** of the fabric body section **110**, where the at least one reinforcement element **130** is configured to reduce sag of the fabric body section **110** when the pipe sling **100** is being lifted with a pipe. It is noted that the pipe sling **100** described above is certified to embody a safe working load (SWL)/rated capacity of 4,500 pounds. The certification is based at least on the following: Federal Specification RR-W-410D, ASME B30.9, API 9A, API 9B, API 2D, the Wire Rope Technical Board's "Wire Rope User's User, the Wire Rope Technical Board's "Wire Rope Sling User's Manual", and the Occupational Safety and Health Administration's "Industrial Slings".

[0027] For the purposes of this disclosure, the term "lifting device" may refer to any device, unit, mechanism, or combination thereof, which is configured and is capable of lifting and moving the pipe when the pipe is housed in any embodiment of pipe sling **100** disclosed. For example, a "lifting device" may include, but is not limited to: a hoist (automatic and/or manual), a crane, and a helicopter.

[0028] It is noted that the configuration of pipe sling **100** disclosed may eliminate the need for end caps for pipes. In the case where fabric body section **110** is longer in length than a pipe wrapped in fabric body section **110**, ends of a pipe are covered and protected by fabric body section **110**.

[0029] In any of the embodiments disclosed, the first set of securement devices **122** and the second set of securement devices **126** include at least four eye to eye nylon lifting straps **123** spaced between at least four ratchet straps **124**.

[0030] In any of the embodiments disclosed, pipe sling **100** includes at least one reinforcement element **130** including at least one metal segment extending along an entirety of the first axis of symmetry **112** from a first end **132** of the fabric body section **110** to a second end **134** of the fabric body section **110**.

[0031] In any of the embodiments disclosed, lifting straps **123** may each comprise two loops (one at each end of strap **123**) for engagement with a lifting device. In another embodiment, lifting straps **123** may each comprise four loops (two at each end of strap **123**) for engagement with a lifting device. In another embodiment, lifting straps **123** may each comprise up to ten loops (five at each end of strap **123**) for engagement with a lifting device. In further embodiments, the entirety of lifting straps **123** and/or the loops comprise a two ply configuration.

[0032] In any of the embodiments disclosed, pipe sling **100** is configured to be utilized with either of a horizontal lifting device or a vertical lifting device.

[0033] In embodiments, pipe sling **100** is configured to support/move/transport a pipe having a maximum length of 12 feet.

[0034] The example systems, methods, and acts described in the embodiments presented previously are illustrative, and, in alternative embodiments, certain acts can be performed in a different order, in parallel with one another, omitted entirely, and/or combined between different example embodiments, and/or certain additional acts can be performed, without departing from the scope and spirit of various embodiments. Accordingly, such alternative embodiments are included in the description herein.

[0035] As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. As used herein, phrases such as

“between X and Y” and “between about X and Y” should be interpreted to include X and Y. As used herein, phrases such as “between about X and Y” mean “between about X and about Y.” As used herein, phrases such as “from about X to Y” mean “from about X to about Y.”

[0036] The above-disclosed embodiments have been presented for purposes of illustration and to enable one of ordinary skill in the art to practice the disclosure, but the disclosure is not intended to be exhaustive or limited to the forms disclosed. Many insubstantial modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The scope of the claims is intended to broadly cover the disclosed embodiments and any such modification. Further, the following clauses represent additional embodiments of the disclosure and should be considered within the scope of the disclosure:

[0037] Clause 1, a pipe sling comprising: a fabric body section, the fabric body section comprising a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section; a first set of securement devices affixed to a first segment of the fabric body section defined by the second axis of symmetry; a second set of securement devices affixed to a second segment of the fabric body section defined by the second axis of symmetry; and at least one reinforcement element positioned along the first axis of symmetry and extending from a first end of the fabric body section to a second end of the fabric body section.

[0038] The pipe sling of Clause 1, wherein the fabric body section comprises nylon canvas fabric.

[0039] The pipe sling of Clause 1, wherein the width of the fabric body section ranges from 24 inches to 36 inches.

[0040] The pipe sling of Clause 1, wherein each of the at least one reinforcement element comprises a metal segment.

[0041] The pipe sling of Clause 1, wherein each of the first set of securement devices and each of the second set of securement devices extend across the width of the fabric body section.

[0042] The pipe sling of Clause 5, wherein each of the first set of securement devices and each of the second set of securement devices are stitched to the fabric body section along the first axis of symmetry.

[0043] The pipe sling of Clause 1, wherein each of the first set of securement devices and each of the second set of securement devices are positioned through a pair of guide loops, each of the pair of guide loops positioned equidistant from the first axis of symmetry.

[0044] The pipe sling of Clause 7, wherein a first guide loop of each of the pair of guide loops of the first set of securement devices and the second set of securement devices are positioned 12 inches from a second guide loop of each of the pair of guide loops of the first set of securement devices and the second set of securement devices.

[0045] The pipe sling of Clause 1, wherein the first set of securement devices and the second set of securement devices each comprise a pair of lifting straps positioned between a pair of ratchet straps.

[0046] The pipe sling of Clause 1, wherein the at least one reinforcement element is configured to reduce sag of the fabric body section when the pipe sling is being lifted with a pipe.

[0047] Clause 11, a system for transporting pipes, comprising: a lifting device; and a pipe sling suspended from the lifting device and wrapped around a diameter of a pipe, the pipe sling comprising: a fabric body section, the fabric body section comprising a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section; a first set of securement devices affixed to a first segment of the fabric body section defined by the second axis of symmetry, wherein a first quantity of the first set of securement devices are secured to the lifting device and a second quantity of the first set of securement devices are wrapped around and secured to the fabric body section and the pipe; a second set of securement devices affixed to a second segment of the fabric body section defined by the second axis of symmetry, wherein a first quantity of the second set of securement devices are

removably affixed to the lifting device and a second quantity of the second set of securement devices are wrapped around and secured to the fabric body section and the pipe; and at least one reinforcement element positioned along the first axis of symmetry and extending from a first end of the fabric body section to a second end of the fabric body section; wherein the at least one reinforcement element is configured to reduce sag of the fabric body section when the pipe sling is wrapped around the pipe while secured to the lifting device.

[0048] Clause 12, the pipe sling of Clause 1, wherein the fabric body section comprises nylon canvas fabric.

[0049] Clause 13, the system of Clause 11, wherein the width of the fabric body section ranges from 24 inches to 36 inches.

[0050] Clause 14, the system of Clause 11, wherein each of the at least one reinforcement element comprises a metal segment.

[0051] Clause 15, the system of Clause 11, wherein each of the first set of securement devices and each of the second set of securement devices extend across the width of the fabric body section.

[0052] Clause 16, the system of Clause 15, wherein each of the first set of securement devices and each of the second set of securement devices are stitched to the fabric body section along the first axis of symmetry.

[0053] Clause 17, the pipe sling of Clause 11, wherein each of the first set of securement devices and each of the second set of securement devices are positioned through a pair of guide loops, each of the pair of guide loops positioned equidistant from the first axis of symmetry.

[0054] Clause 18, the pipe sling of Clause 17, wherein a first guide loop of each of the pair of guide loops of the first set of securement devices and the second set of securement devices are positioned 12 inches from a second guide loop of each of the pair of guide loops of the first set of securement devices and the second set of securement devices.

[0055] Clause 19, the pipe sling of Clause 11, wherein the first set of securement devices and the second set of securement devices each comprise a pair of lifting straps positioned between a pair of ratchet straps.

[0056] Clause 20, a pipe sling for supporting a pipe, comprising: a nylon canvas fabric body section, the fabric body section comprising a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section, the width of the fabric body section ranging from 24 inches to 36 inches; a first set of securement devices affixed to a first segment of the fabric body section defined by the second axis of symmetry and a second set of securement devices affixed to a second segment of the fabric body section defined by the second axis of symmetry, each of the first set of securement devices and each of the second set of securement devices extending across the width of the fabric body section and stitched to the fabric body section along the first axis of symmetry, the first set of securement devices and the second set of securement devices including at least four eye to eye nylon lifting straps spaced between at least four ratchet straps; and at least one reinforcement element comprising at least one metal segment extending along an entirety of the first axis of symmetry from a first end of the fabric body section to a second end of the fabric body section, the at least one reinforcement element configured to reduce sag of the fabric body section when the pipe sling is being lifted with a pipe; wherein a safe working load (SWL) of the pipe sling, when lifting the pipe, ranges from 3,000 pounds to 4,500 pounds.

Claims

1. A pipe sling comprising: a fabric body section, the fabric body section comprising a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section; a first set of securement devices affixed to a first segment of the fabric body section defined by the second axis of symmetry; a second set of

securement devices affixed to a second segment of the fabric body section defined by the second axis of symmetry; and at least one reinforcement element positioned along the first axis of symmetry and extending from a first end of the fabric body section to a second end of the fabric body section.

2. The pipe sling of claim 1, wherein the fabric body section comprises nylon canvas fabric.

3. The pipe sling of claim 1, wherein the width of the fabric body section ranges from 24 inches to 36 inches.

4. The pipe sling of claim 1, wherein each of the at least one reinforcement element comprises a metal segment.

5. The pipe sling of claim 1, wherein each of the first set of securement devices and each of the second set of securement devices extend across the width of the fabric body section.

6. The pipe sling of claim 5, wherein each of the first set of securement devices and each of the second set of securement devices are stitched to the fabric body section along the first axis of symmetry.

7. The pipe sling of claim 1, wherein each of the first set of securement devices and each of the second set of securement devices are positioned through a pair of guide loops, each of the pair of guide loops positioned equidistant from the first axis of symmetry.

8. The pipe sling of claim 7, wherein a first guide loop of each of the pair of guide loops of the first set of securement devices and the second set of securement devices are positioned 12 inches from a second guide loop of each of the pair of guide loops of the first set of securement devices and the second set of securement devices.

9. The pipe sling of claim 1, wherein the first set of securement devices and the second set of securement devices each comprise a pair of lifting straps positioned between a pair of ratchet straps.

10. The pipe sling of claim 1, wherein the at least one reinforcement element is configured to reduce sag of the fabric body section when the pipe sling is being lifted with a pipe.

11. A system for transporting pipes, comprising: a lifting device; and a pipe sling suspended from the lifting device and wrapped around a diameter of a pipe, the pipe sling comprising: a fabric body section, the fabric body section comprising a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section; a first set of securement devices affixed to a first segment of the fabric body section defined by the second axis of symmetry, wherein a first quantity of the first set of securement devices are secured to the lifting device and a second quantity of the first set of securement devices are wrapped around and secured to the fabric body section and the pipe; a second set of securement devices affixed to a second segment of the fabric body section defined by the second axis of symmetry, wherein a first quantity of the second set of securement devices are removably affixed to the lifting device and a second quantity of the second set of securement devices are wrapped around and secured to the fabric body section and the pipe; and at least one reinforcement element positioned along the first axis of symmetry and extending from a first end of the fabric body section to a second end of the fabric body section; wherein the at least one reinforcement element is configured to reduce sag of the fabric body section when the pipe sling is wrapped around the pipe while secured to the lifting device.

12. The pipe sling of claim 1, wherein the fabric body section comprises nylon canvas fabric.

13. The system of claim 11, wherein the width of the fabric body section ranges from 24 inches to 36 inches.

14. The system of claim 11, wherein each of the at least one reinforcement element comprises a metal segment.

15. The system of claim 11, wherein each of the first set of securement devices and each of the second set of securement devices extend across the width of the fabric body section.

16. The system of claim 15, wherein each of the first set of securement devices and each of the

second set of securement devices are stitched to the fabric body section along the first axis of symmetry.

17. The pipe sling of claim 11, wherein each of the first set of securement devices and each of the second set of securement devices are positioned through a pair of guide loops, each of the pair of guide loops positioned equidistant from the first axis of symmetry.

18. The pipe sling of claim 17, wherein a first guide loop of each of the pair of guide loops of the first set of securement devices and the second set of securement devices are positioned 12 inches from a second guide loop of each of the pair of guide loops of the first set of securement devices and the second set of securement devices.

19. The pipe sling of claim 11, wherein the first set of securement devices and the second set of securement devices each comprise a pair of lifting straps positioned between a pair of ratchet straps.

20. A pipe sling for supporting a pipe, comprising: a nylon canvas fabric body section, the fabric body section comprising a first axis of symmetry extending along a length of the fabric body section and a second axis of symmetry extending along a width of the fabric body section, the width of the fabric body section ranging from 24 inches to 36 inches; a first set of securement devices affixed to a first segment of the fabric body section defined by the second axis of symmetry and a second set of securement devices affixed to a second segment of the fabric body section defined by the second axis of symmetry, each of the first set of securement devices and each of the second set of securement devices extending across the width of the fabric body section and stitched to the fabric body section along the first axis of symmetry, the first set of securement devices and the second set of securement devices including at least four eye to eye nylon lifting straps spaced between at least four ratchet straps; and at least one reinforcement element comprising at least one metal segment extending along an entirety of the first axis of symmetry from a first end of the fabric body section to a second end of the fabric body section, the at least one reinforcement element configured to reduce sag of the fabric body section when the pipe sling is being lifted with a pipe; wherein a safe working load (SWL) of the pipe sling, when lifting the pipe, ranges from 3,000 pounds to 4,500 pounds.
